Martin Farkas

PhD Candidate · Computer Engineering
Critical Systems Research Group (FTSRG), BME
Trust & Privacy Engineer · Applied Security Researcher

martin.farkas@edu.bme.hu · BlackLight54 · Martin Farkas
Martin Farkas · 0000-0003-4260-717X · home.mit.bme.hu/~farkas · MTMT: 10098507
WoS: MBH-7169-2025 · SC Scopus: 59515829300 · DBLP: 393/6532 · doktori.hu: 54225

About

Martin Farkas is a PhD student in the Critical Systems Research Group at the Budapest University of Technology and Economics, where he completed his MSc with honors in 2025. His research is on the **model-driven design of decentralized trust solutions**: applying formal modeling and verification techniques to digital identity systems built on Self-Sovereign Identity standards and Zero-Knowledge Proofs. The aim is to make trust assumptions in such systems explicit and machine-checkable, rather than implicit and audited after the fact.

The dissertation pursues three threads: (i) *cryptographically verifiable policy evaluation*, where rules can be checked without revealing the data they decide on; (ii) *decentralized, secure, dynamic, privacy-preserving business process collaborations* where each participant proves correctness without a central coordinator; and (iii) *model-driven design tooling for novel decentralized trust solutions*.

Research Interests

Formal verification of decentralized trust systems · Self-Sovereign Identity (SSI) and Verifiable Credentials · Zero-Knowledge Proofs and privacy-preserving cryptographic protocols · Hyperledger Fabric and permissioned distributed ledgers · model-driven engineering for digital identity · business process modeling (BPMN) and security analysis.

Education

PhD in Computer Engineering — Budapest, Hungary

Budapest University of Technology and Economics — 2025 — Present

- Topic: *Model-driven design of decentralized trust solutions*
- Critical Systems Research Group (FTSRG)
- Department of Artificial Intelligence and Systems Engineering (MIT)
- Supervisor: Dr. Imre Kocsis

MSc in Computer Engineering — Budapest, Hungary

Budapest University of Technology and Economics — 2023 — 2025

- Specialization: IT Security, Critical Systems
- Final grade: **Summa Cum Laude**
- Thesis: *Self-Sovereign Identity based Self-Evaluated Policies*

BSc in Computer Engineering — Budapest, Hungary

Budapest University of Technology and Economics — 2019 — 2023

- Specialization: System Engineering
- Thesis: *Self-Sovereign Identity supported payment on the openCBDC platform*

Secondary School — Győr, Hungary

Révai Miklós Gimnázium és Kollégium · 6-year programme — 2013 — 2019

Academic Positions

Research Assistant — Budapest, Hungary

Critical Systems Research Group (FTSRG), BME — 2023 — Present

- Privacy-preserving protocols for blockchain and SSI systems
- Zero-Knowledge Proof systems for verifiable credentials
- Course delivery in Self-Sovereign Identity, distributed systems, and blockchain

- Mentoring BSc and MSc students on SSI and blockchain research

Publications

A Self-Orchestration Model for Business Collaborations with Verifiable Process History Credentials — Seville, Spain

Farkas M., Péter B.Z., Kocsis I. · BPM 2025 Forum · LNBIP 564, Springer · doi:10.1007/978-3-032-02929-4_4 — Aug 2025

A Prolog-based Approach to Self-Evaluated, Declarative and Zero-Knowledge Verifiable Policies — Krakow, Poland

Farkas M., Toldi B.Á., Péter B.Z., Kocsis I. · EuroCyberSec Workshop · MASCOTS 2024, IEEE — Oct 2024

Design Space Exploration of Verifiable Credential Schemas using Partial Graph Modeling — Szeged, Hungary

Farkas M., Péter B.Z., Kocsis I. · CS² · 14th Conference of PhD Students in Computer Science — Jul 2024

Service

Sub-reviewer · ICSE 2026 (Demonstrations track) — for Grace Lewis · Nov 2025

Sub-reviewer · SAC 2026 — for Imre Kocsis · Nov 2025

Sub-reviewer · SafeComp 2026 — for Zoltán Micskei · Apr 2026

Awards & Honours

3rd place — National Scientific Students' Conference (OTDK) — Cybersecurity section

Self-evaluated policies using Zero-Knowledge Proofs — Apr 2025

3rd place — BME VIK Students' Scientific Conference (TDK) — System Modelling section

Design Space Exploration of Verifiable Credential Schemas using Partial Graph Modeling — Nov 2024

1st place — BME VIK Students' Scientific Conference (TDK) — Information Systems section

Self-evaluated policies using Zero-Knowledge proofs — Nov 2023

Special prize — BME VIK Students' Scientific Conference (TDK) — Information Systems section

Payments in openCBDC with Self-Sovereign Identities — from the verifiable to the private — Nov 2022

Summa Cum Laude

MSc in Computer Engineering, BME — 2025

Research Projects

MNB–BME CBDC Cooperation — Budapest, Hungary

Hungarian National Bank & BME — 2024 — 2025

Integrated an SSI-based Zero-Knowledge Proof protocol into a Central Bank Digital Currency prototype, extending earlier work in the Critical Systems Research Group.

EMAP — Eseményalapú Adatszolgáltatási Platform — Budapest, Hungary

BME · NAV & Hungarian Treasury · ET 3.1 — Sep 2025 — Present

Designing a graph-based change impact analysis methodology for EMAP, a nationwide event-driven public-sector data platform serving Hungarian state organizations. The approach adapts Error Propagation Analysis to requirements management over a typed traceability graph.

DC4EU — Digital Credentials for Europe — BME

EU large-scale Verifiable Credentials pilot · Grant Agreement 101102611 — 2025

Facilitating BME's contribution to the EU large-scale pilot for Verifiable Credentials across the education and social-security domains.

DigitalTech EDIH 1.0 & 2.0 — Blockchain task at BME

European Digital Innovation Hubs initiative · European Commission — 2023 — Present

Coursework development on permissioned blockchains and digital identity (SSI) for EDIH 1.0 (completed) and EDIH 2.0 (ongoing).

Training SMEs for the Digital Decade (SME4DD)

EU SME4DD project — 2023 — Present

Coursework development for the digital identity management module of the EU SME4DD training programme.

Industry Experience

Quanopt Kft. — Budapest, Hungary

Engineering & research consulting — 2023 — Present

- *Lead Engineer & Crypto Engineer*, Summer 2025 — IoP/Morpheus ⟶ Hyperledger Fabric port, in cooperation with Morpheus.tech. Reimplemented Morpheus's DID state machine and credential operations as Fabric chaincode, replacing Hydra's Layer-2 block-event model with Fabric's endorsement flow.
- *Systems Engineer*, 2023 — Lead developer on a Hyperledger Fabric chaincode for metering-data storage at a Hungarian gas provider; Fablo and Java for the chaincode, Node.js/TypeScript for the Swagger REST API layer, plus the performance measurement campaign.

Blockchain Node Engineer — Budapest, Hungary

NAS Kft. — 2022

- Built and customized a Go Ethereum (Geth) node for a GPU-based mining platform
- Conducted candidate mining-software review and market analysis informing platform design choices

Teaching & Supervision

BSc thesis & TDK supervision (BME VIK)

Cloud-ready, compositionally verifiable zero-knowledge evaluation of declarative policies — 2025

TDK supervision (BME VIK)

LLM-assisted creation and refinement of error propagation analysis models — 2025

Student mentoring

BSc and MSc students · SSI, distributed systems, blockchain — 2023 — Present

Supervising student research projects on Zero-Knowledge Proof learning paths and DeFi/blockchain topics.

Selective Programs

Marktoberdorf Summer School (MOD25) — Marktoberdorf, Germany

Specification and verification of secure cyberspaces — Aug 2025

Conferences Attended

BSidesBUD 2026 — Budapest, Hungary

10th anniversary edition · community-driven IT security conference, Central Europe — Apr 2026

Hacktivity 2025 — Budapest, Hungary

Central & Eastern Europe's longest-running IT security festival — 2025

HUSTEF 2024 — Budapest, Hungary

Hungarian Software Testing Forum — 2024

ITBN Conf-Expo 2024 — Budapest, Hungary

value chAIned · Informatikai Biztonság Napja · Hungary's IT security conference — Sep 2024

Technical Skills

Formal Methods	**Tamarin Prover**, **Refinery**, epistemic logic, model-driven engineering, SysML, UML, PlantUML

Cryptography	**Zero-Knowledge Proofs**, Circom, Halo2, snark.js, zkVMs, Hyperledger AnonCreds, W3C Verifiable Credentials, DIDComm
Blockchain & DLT	**Hyperledger Fabric**, Hyperledger Aries, Hyperledger Indy, Go Ethereum, Solidity, Truffle, Ganache
Programming	**Rust**, **TypeScript**, **Python**, Java, Kotlin, C, C++, C#, JavaScript, Solidity, Prolog, Circom, Elixir, Bash
Web & Backend	Node.js, NestJS, Express, Next.js, Spring, Tailwind CSS, REST/Swagger
DevOps & Data	Docker, Kubernetes, Git, GitHub Actions, Jenkins, SQL, MongoDB
Languages	**Hungarian (Native)**, **English (C1, IELTS 8.0)**

Referees

Dr. Imre Kocsis

Associate Professor, Department of Artificial Intelligence and Systems Engineering, BME · Critical Systems Research Group · *PhD advisor*

Dr. Zoltán Micskei

Associate Professor, head of the Critical Systems Research Group, BME

Dr. László Gönczy

Associate Professor and Head of the Department of Artificial Intelligence and Systems Engineering, BME · former supervisor at Quanopt Kft.

Dr. Oszkár Semeráth

Research Fellow, Critical Systems Research Group, BME · co-developer of Refinery

Prof. András Pataricza

Professor Emeritus, Department of Artificial Intelligence and Systems Engineering, BME · founder of the Fault Tolerant Systems Research Group (1994)